ELECTRICAL—RAIN SENSE WIPER SYSTEM INFORMATION—DIAGNOSTIC AID—SERVICE TIPS	Article No. 02-4-4

LINCOLN: 2000-2002 LS

ISSUE

This information is a supplement to the Rain Sensor Wiper System information provided in the Lincoln LS Workshop Manual. This Service Information is intended as an aid to assist in the diagnosis and repair of Rain Sensor Wiper System conditions.

ACTION

Refer to the following Service Tips to help repair Rain Sensor Wiper System conditions.

SERVICE INFORMATION

NOTE

THIS TSB IS INTENDED TO HELP IDENTIFY AND DIAGNOSE CUSTOMER COMPLAINTS WITH THE RAIN SENSE WIPER SYSTEM. THIS TSB SHOULD NOT BE USED AS A SUBSTITUTE BUT RATHER AS A ENHANCEMENT TO THE WORKSHOP MANUAL. THE PIN POINT TESTS IN THE WORKSHOP MANUAL SHOULD BE PERFORMED BEFORE USING THIS TSB.

NOTE

CHECK BASIC WIPER SYSTEM OPERATION BEFORE ATTEMPTING TO DIAGNOSE THE RAIN SENSE WIPER SYSTEM. IF CONCERNS ARE DISCOVERED WITH BASIC WIPER OPERATION, REPAIRS MUST BE MADE TO RESTORE THE BASIC SYSTEM BEFORE ATTEMPTING TO EVALUATE THE RAIN SENSE WIPER SYSTEM.

Principles of Operation:

The Rain Sense Wiper System feature automatically operates the wipers when moisture accumulates on the windshield. The system is comprised of three (3) components: Multi-Function Switch (MFS), Front Electronics Module (FEM) and the Rain Sense Module (RSM). In order to detect rain on the windshield the RSM directs a light beam through one of its silicone pads (emitter) into the windshield. The light beam bounces on the windshield and weakens if there is presence of moisture. At the end of its travel distance the light beam is returned from the windshield through the opposite side silicone pad (receiver) on the RSM. The remaining intensity of the light beam is then measured by the RSM. If the drop in light beam signal is sufficient enough the RSM will direct the FEM to operate the wipers. In order to accurately determine the speed in which the wipers should operate the RSM continuously measures and analyzes the light beams intensity to determine the amount of moisture that is present (Figure 1). Depending on the intensity of the rain the RSM will command the wipers to operate in one of three modes: Interval, Low Speed and High Speed. The RSM is continuously calibrating itself to accommodate changes in ambient light and temperature since these outside factors will effect system operation.

There are four (4) wires involved in the operation of the Rain Sense Wiper System:

1. Mode Line (pin 1 at RSM, circuit 8-KA41) The Mode Line is a signal voltage input to the FEM from the RSM to operate the Rain Sense Wipers. The MFS must be in the correct positions for Rain Sense Operation (Figure 2).
2. Delay Line (pin 3 at RSM, circuit 10-KA41) The Delay Line is a MFS input to the RSM, which tells the RSM what position the MFS is selected. This input stages the RSM to manipulate the voltage on the Mode Line if rain is present.
3. Ground Line (pin 4 at RSM, circuit 7-KA41).
4. RSM Supply Voltage (pin 6 at RSM, circuit 29-KA41).

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The voltage on the Delay Line will be approximately 7.3 volts when the MFS is in the AUTO position. The voltage on the Mode Line will be approximately 4.5 volts in the INTERMITTENT position. When the RSM detects enough moisture for a wipe event it will drop the voltage on the Mode Line to approximately 2.4 volts. Consequently, this is the same voltage the Mode Line will have when the MFS is in low speed or single wipe position during non-Rain Sense operation. When the FEM receives the drop of voltage to 2.4 volts on the Mode Line it will operate the wiper relay to swipe the wipers. Voltage observed on the Mode & Delay lines will vary slightly depending on battery condition and if the engine is running or off during the test.

The following may be examples of customer concerns on the Rain Sense Wiper System:

1. System Operates Erratically - Some customers may complain that the system operates erratically, or changes speed often. It is normal for the Rain Sense Wipers to change speed (low-high) and frequency (interval) when the system is active. The system is constantly monitoring the amount of moisture present on the RSM sensitive section of the windshield and operates the wipers accordingly. Depending on the wiping characteristics desired by the driver the MFS may need to be manually changed to a different speed.
2. System Inoperative When Windshield Is Frosted - A frosted windshield will not allow the light beam to get through the glass for the RSM to make a observation of possible moisture. If the RSM detects frost during the initial wipe it shuts down operation for 15 minutes. During the 15 minute delay, the sensor does not react if water hits the windshield. After 15 minutes have elapsed, another wipe is commanded and the RSM may now begin normal operation if the frost has cleared. If the windshield is still frosted, the RSM will repeat the 15 minute waiting period. If the same condition is detected for the third time, the RSM shuts down completely until the next key off/on event. A cracked windshield may exhibit the symptoms.
3. System Shuts Down During Heavy Rain Fall - The RSM may operate in the same fashion as a frosted windshield event during heavy rain (15 minute delay). The extreme blanket of rain over the RSM will not allow the system to operate correctly causing it to go into a 15 minute delay period. During such an event the wiper system must be turned on manually using the MFS. This should be considered as normal operation.
4. System Intermittently Cycles The Wipers One Time During Engine Start - If the system consistently left in an active state (MFS in the INT & AUTO positions), the wipers may cycle one time during a engine start event. This is a normal condition that may be referenced in the Owner's Guide under "Moisture Sensitive Wipers".
5. System Is Slow To Start (Slow To Detect Rain) - The system may be slow to react to rain if it is consistently left in an active state (MFS in the AUTO & INT positions). If operating the vehicle in dry conditions with the system in an active state, or if the system is activated with the MFS and no rain is present for 1 minute, the system will go into Power Save or Stand By Mode. In this mode, the system will shut off for one second in 50 millisecond intervals. The RSM will only sense for moisture during the 50 millisecond intervals. If no rain is sensed the system will continue shutting off in one second intervals. If rain should fall during this one second interval, the system will remain off until the one second time window has lapsed. When rain is sensed the system will become active and operate accordingly (Figure 3).
6. System Has An Occasional "Phantom" Wipe - If the driver reports an intermittent Phantom wipe while driving when the system is in an active state, proceed to the Visual Checks section of this TSB.
7. System Is Completely Inoperative - Verify that all basic wiper system operations are in order. Service the basic wiper system before attempting to diagnose the Rain Sense Wiper System. If the system is verified as completely inoperative, proceed to the Visual Checks section of this TSB.

Visual Checks:

- If windshield has been replaced during service check that both Silicone Pads are present on the RSM. If one or both of the silicone pads are missing, replace the RSM. The pads are not available separately from the RSM. (Figure 4).
- Before attempting to diagnose the Rain Sense Wiper System, be sure that the driver is placing the MFS in both the AUTO and INTERMITTENT position. Refer to appropriate page the Owner's Guide to verify correct MFS position. (Figure 2).
- Make certain no dirt, ink or window tinting is covering the RSM area of the windshield. If any ink or tint is covering the RSM area, or blocking any portion of the RSM silicone pads, replace the windshield. Correct placement of the RSM may be inspected from outside the vehicle. (Figure 4).
- Check for cracks in the windshield. Cracks in the windshield affect the working parameters of the RSM even if the crack is away from the RSM area. The windshield should be replaced with the correct level glass and then evaluate Rain Sense Wiper System performance.
- Check if the windshield has been replaced during service. RSM attaching bracket location on the windshield is critical. Do not attempt to remove the RSM brackets from the original windshield and reattach RSM brackets to a new windshield. The correct windshield must be installed in the vehicle if the Rain Sense Wiper System is expected to operate properly. Carlite and most other windshield manufacturers have windshields available for specific RSM applications.

NOTE

IF THE RSM BRACKETS ON THE WINDSHIELD WERE ATTACHED BY ANY OTHER MEANS THAN THE ORIGINAL MANUFACTURER THE WINDSHIELD MUST BE REPLACED BEFORE EVALUATING RAIN SENSE WIPER SYSTEM PERFORMANCE.

- Remove the RSM from the windshield. Measure the distance between the two brackets at the top and bottom (parallelism). Measure the gap between the ink opening and the brackets for even spacing on each side. Replace the windshield if the RSM is not correctly centered in the ink opening or if the brackets are out of parallel. (Figure 5).
- Use a business card or razor blade to check RSM bracket flushness and uniform attachment to the windshield. If a gap in the adhesive is detected between the glass and RSM bracket, or if it appears that the RSM bracket is not flush to the windshield, the windshield must be replaced. Do not attempt to remedy any attachment or flushness issues the RSM brackets.
- Remove the rain sensor by pulling the green tabs outward. Visually check the two (2) clear silicone rubber pads on the RSM. These pads must be clear in appearance. If the pads appear to be cloudy or spotted, replace the RSM. If there is presence of dirt or dust on the pads, use transparent tape (Scotch Tape) to remove the particles. It is not unusual for the pads to stick on the windshield when removing the RSM. If attempting to re-install the pads on the RSM, make sure your hands are clean. Replacing the RSM may be wise if there are any complications. Re-install the RSM and evaluate the Rain Sense Wiper System performance.
- Do not attempt to diagnose the system by observing the operation of the LED visible on the RSM. It is normal for the LED to lose illumination at times. It is also normal for the LED to flash at times.
- If the Rain Sense Wiper System is still inoperative, perform the following tests.

NOTE

MOST ISSUES WITH RAIN SENSE WIPER SYSTEM HAVE BEEN ATTRIBUTED TO A RSM ATTACHMENT CONCERN OR A RESULT OF AN INCORRECT WINDSHIELD INSTALLED DURING SERVICE.

Pin Point Tests:

1. Does the INTERMITTENT wiper function operate with normal MFS command?
   - No - Diagnose the wiper system using normal Workshop Manual diagnostics. Resistance across the MFS, pin 1 and 2, at the MFS connector should be 1910 ohms. Refer to Workshop Manual Section 501-16 for addition information.
   - Yes - This proves that the MFS is able to pull the voltage down on the Delay Line to the correct voltage (7.3 volts) and that the FEM is capable of reading the voltage and acting accordingly. This also proves that the delay line is not shorted to power or ground. Continue to Step 2.

2. Does the Single Wipe function operate normally?
   - No - Diagnose the wiper system using normal Workshop Manual Diagnostics. Refer to Workshop Manual Section 501-16 for additional information.
   - Yes - This proves that the FEM is capable of operating the wipers accordingly when the Mode Line is pulled down to approximately 2.4 volts. This also proves that the Mode Line is not shorted to power or ground. Continue to Step 3.
3. Electrically disconnect the RSM and check C914, pin 6 for 12v, KOEO.
   - No - Repair the circuit from fuse 216 in the Central Junction Box to the RSM, inspecting C311 on the passenger side C-Pillar as a possible cause of the fault. Refer to EV/TM, Section 81-5 for addition information.
   - Yes - Continue to Step 4.
4. RSM electrically disconnected, place the MFS in INTERMITTENT position and KOEO, check for approximately 7.3 volts on the Delay Line, C914 pin 3.
   - No - Repair the open or high resistance in the circuit. Refer to EV/TM, Section 81-5 for addition information.
   - Yes - Continue to Step 5.
5. RSM electrically disconnected, MFS in AUTO position and KOEO, check for approximately 4.5 volts on the Mode Line, C914 pin 1.
   - No - Disconnect the MFS and check resistance across pins 2 and 3 at the MFS connector. Resistance measured should be about 2675 ohms. If the resistance value is acceptable, locate and repair the open circuit or high resistance on the Mode Line. Refer to EV/TM, Section 81-5 for addition information.
   - Yes - Replace with new or known good RSM.

PART NUMBER	PART NAME
XW4Z-5403100-BA	Windshield - All 2000 Model Year
1W4Z-5403100-BB	Windshield - All 2001 Model Year To Build Date 7/16/01
2W4Z-5403100-BA	Windshield - 2001 Model Built From 7/16/01 And All 2002 Model Year
2W4Z-13K359-AA	Multifunction Switch
XW4Z-17D547-AA	Rain Sense Module
2W4Z-17D550-AA	Rain Sense Module Cover

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 102000, 202000

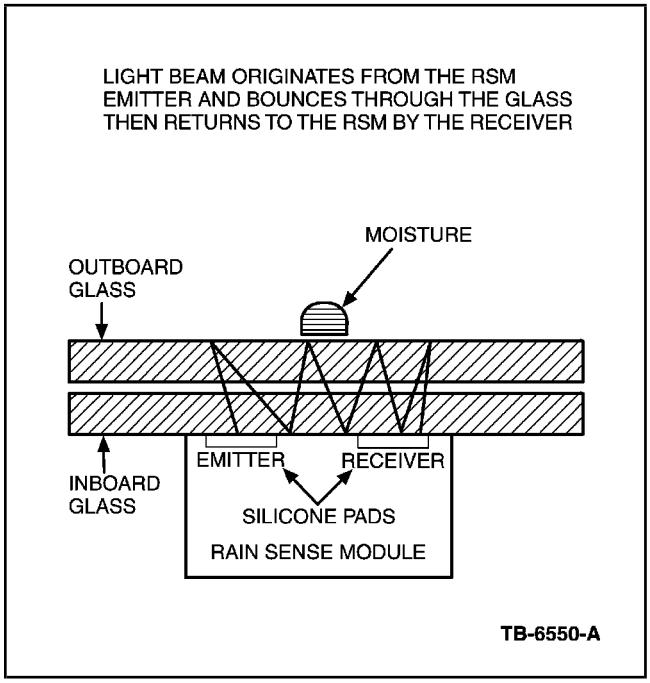


Figure 1 - Article 02-4-4

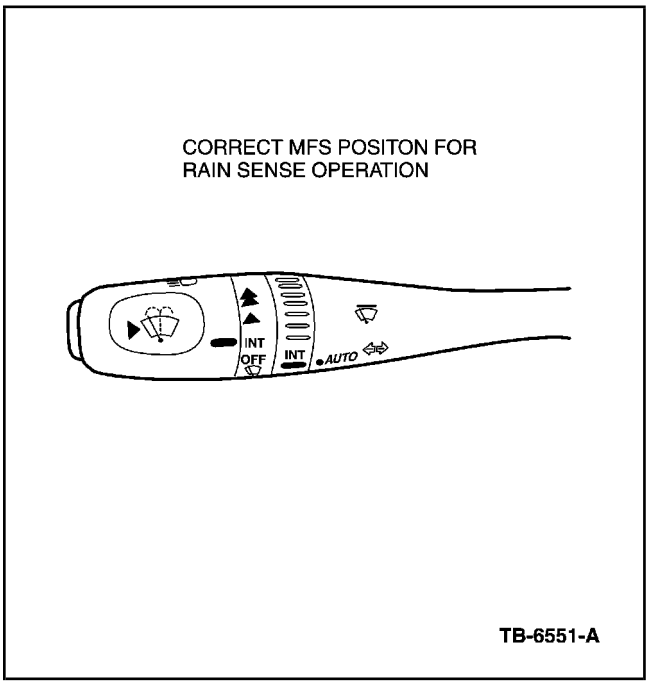


Figure 2 - Article 02-4-4

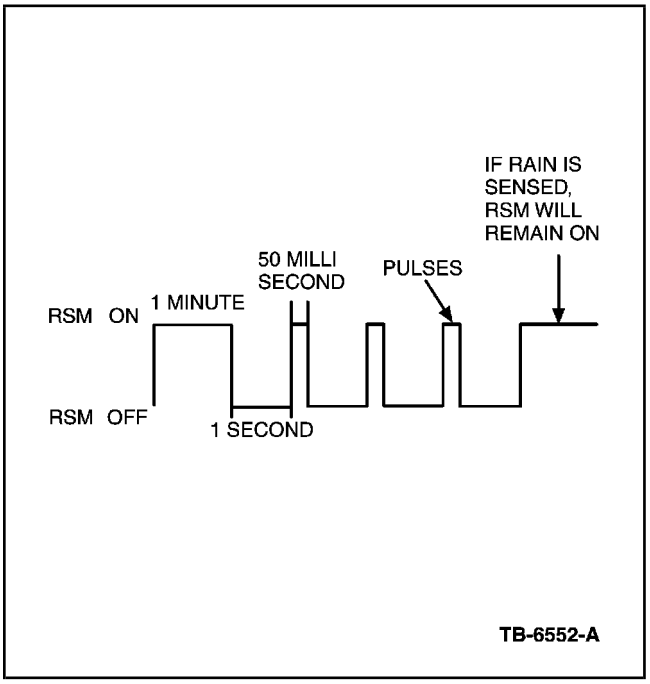


Figure 3 - Article 02-4-4

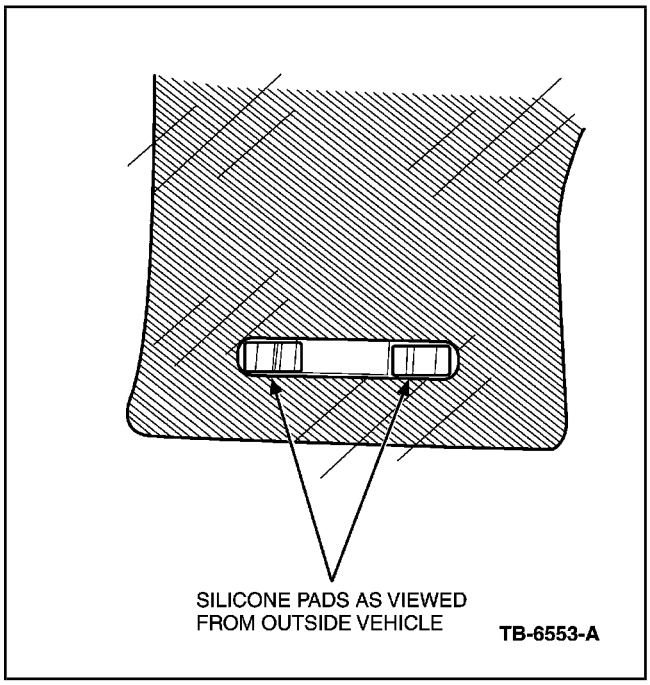


Figure 4 - Article 02-4-4

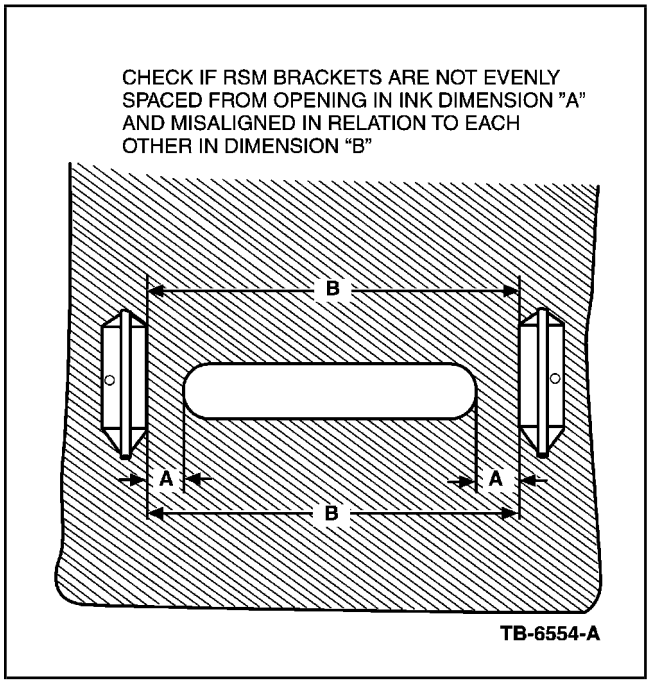


Figure 5 - Article 02-4-4